

Heaven's Light is Our Guide
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Lab Report 6:
Design & Analysis of a Low-Pass FIR and IIR Filter & Their frequency Response Comparisons

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Design & Analysis of a Low-Pass FIR and IIR Filter & Their frequency Response Comparisons

Theory

This experiment compares a windowed FIR low-pass filter and a Butterworth IIR low-pass filter by designing both to attenuate frequencies above a specified cutoff and by inspecting their frequency- and time-domain behaviour [1, 2].

Design parameters

- Sampling frequency: f_s .
- Cutoff frequency: f_c .
- FIR order: N (even or odd – the design used a fixed order).
- IIR prototype: Butterworth of order M (maximally flat in the passband).

The normalized digital cutoff used for design (normalized to Nyquist, as used by many DSP tool-boxes) is

$$\omega_n = \frac{f_c}{f_s/2} \quad (0 < \omega_n \leq 1).$$

The corresponding angular cutoff in radians/sample is

$$\Omega_c = 2\pi \frac{f_c}{f_s}.$$

FIR design (windowed FIR)

The ideal (infinite-length) low-pass impulse response with cutoff Ω_c is

$$h_d[n] = \begin{cases} \frac{\sin(\Omega_c(n - N/2))}{\pi(n - N/2)}, & n \neq N/2, \\ \frac{\Omega_c}{\pi}, & n = N/2. \end{cases}$$

A causal, finite-length FIR is obtained by windowing:

$$h[n] = h_d[n] w[n], \quad n = 0, \dots, N,$$

where $w[n]$ is a window (e.g. Hamming, Hann, Blackman). The resulting frequency response is

$$H(e^{j\Omega}) = \sum_{n=0}^N h[n] e^{-j\Omega n}.$$

For symmetric coefficients (linear phase) the group delay is constant and approximately

$$\tau_g \approx \frac{N}{2} \text{ samples.}$$

Window design and the trade-offs between main-lobe width and sidelobe level are discussed in detail in standard texts [1, 3].

IIR design (Butterworth)

The (continuous-time) Butterworth prototype magnitude-squared response is

$$|H_a(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2M}},$$

where M is the analog prototype order and Ω_c is the analog cutoff. A digital Butterworth is typically obtained by prewarping and bilinear transform:

$$s \leftarrow \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}, \quad T = \frac{1}{f_s},$$

leading to a causal IIR with difference equation (direct form)

$$y[n] = \frac{1}{a_0} \left(\sum_{k=0}^M b_k x[n - k] - \sum_{k=1}^M a_k y[n - k] \right),$$

and frequency response

$$H(e^{j\Omega}) = \frac{\sum_{k=0}^M b_k e^{-j\Omega k}}{\sum_{k=0}^M a_k e^{-j\Omega k}}.$$

Butterworth IIRs give steep roll-off for modest orders but exhibit nonlinear phase; design procedures and bilinear-transform details are treated in standard DSP references [2, 1].

Frequency-domain measures

Magnitude in decibels is plotted as

$$\text{Mag}_{\text{dB}}(\Omega) = 20 \log_{10} |H(e^{j\Omega})|.$$

When computing the dB scale numerically, avoid log of zero by flooring small magnitudes (e.g. $\max(|H|, \epsilon)$). Numerical plotting and display considerations are discussed in practical DSP texts [3].

Time-domain considerations

Filter impulse response length (finite for FIR, infinite for IIR) determines transient behaviour. The FIR's linear phase preserves waveform shape aside from the constant delay τ_g ; the IIR can introduce phase distortion and ringing near transients [2, 1].

Examples

1. Example A (typical audio-ish design)

- $f_s = 48 \text{ kHz}$, $f_c = 5 \text{ kHz}$.

- FIR order $N = 50$ (use Hamming window). Normalized cutoff for toolbox calls:

$$\omega_n = \frac{5000}{48000/2} = 0.20833.$$

MATLAB (conceptual):

```
Wn = 5000/(48000/2); b = fir1(50, Wn, hamming(51));
```

- Butterworth IIR order $M = 4$ (prewarp + bilinear):

```
[b,a] = butter(4, Wn); y = filter(b,a,x);
```

2. Example B (narrow transition)

- $f_s = 32$ kHz, $f_c = 2$ kHz, FIR order $N = 200$ to obtain a sharper transition; compare to a Butterworth $M = 6$.
- Expect: FIR has clean linear-phase passband, large group delay ≈ 100 samples; Butterworth attains similar attenuation with lower order but nonlinear phase.

Practical notes and interpretation

- Use sufficiently dense frequency sampling (e.g. several hundred or thousand points) to capture detailed response features.
- Handle numerical zeros when converting to dB and when plotting narrow stop-bands.
- For time-domain fidelity (no phase distortion) prefer FIR; for low-complexity steep roll-off prefer IIR and consider phase compensation if needed.

The two filter families each have trade-offs: FIR yields linear phase and predictable group delay at the cost of higher order for sharp transitions, while Butterworth IIR achieves sharp attenuation with lower order but exhibits nonlinear phase and infinite-duration impulse response. The frequency- and time-domain plots provide direct visualization of these trade-offs for the chosen parameters[1, 2, 3].

Used Tools

- Matlab Online
- LaTeX

Code & Output

Code

```

1 fs = 1000;
2 fcut = 100;
3 n = 50;
4
5 % FIR Filter
6 b_fir = fir1(n, fcut/(fs/2));
7
8 % IIR Filter (Butterworth)
9 [b_iir, a_iir] = butter(4, fcut/(fs/2));
10
11 % Frequency responses (use freqz to get
12   ↪ data, then plot in subplots)
13 Nfreq = 512;
14 [Hf, wf] = freqz(b_fir, 1, Nfreq, fs);
15 [Hi, wi] = freqz(b_iir, a_iir, Nfreq, fs);
16
17 figure;
18 subplot(3,1,1);
```

```

18 plot(wf, 20*log10(abs(Hf)));
19 grid on;
20 title('FIR Magnitude Response');
21 xlabel('Frequency (Hz)');
22 ylabel('Magnitude (dB)');
23
24 subplot(3,1,2);
25 plot(wi, 20*log10(abs(Hi)));
26 grid on;
27 title('IIR Magnitude Response');
28 xlabel('Frequency (Hz)');
29 ylabel('Magnitude (dB)');
30
31 % Compare outputs (time domain)
32 t = 0:1/fs:1;
33 x = sin(2*pi*50*t) + 0.5*randn(size(t));
34 y_fir = filter(b_fir, 1, x);
35 y_iir = filter(b_iir, a_iir, x);
36
37 subplot(3,1,3);
38 plot(t, x, 'k', t, y_fir, 'r', t, y_iir,
    ↪ 'b');
39 grid on;
40 legend('Original', 'FIR Output', 'IIR
    ↪ Output');
41 xlabel('Time (s)');
42 ylabel('Amplitude');
43 title('Time-domain Signals');

```

Output

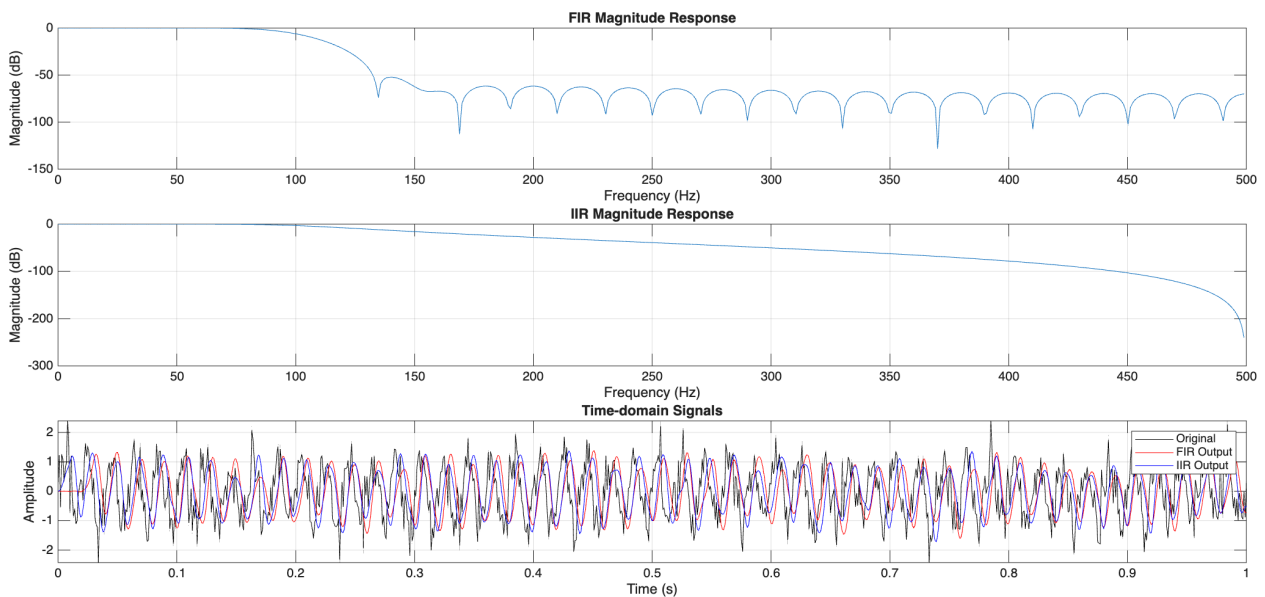


Figure 1: Frequency and Time Domain Responses of FIR and IIR Filters

Analysis

Brief analysis — FIR vs IIR low-pass filters (based on Figure 1)

Summary

- Top subplot (FIR magnitude, dB): clear passband near 0 Hz, a transition band, then periodic deep nulls / stop-band ripples at higher frequencies; passband is fairly flat near 0 dB.
- Middle subplot (IIR / Butterworth magnitude, dB): smooth monotonic roll-off from the passband to deep attenuation without periodic nulls; the curve decreases smoothly and reaches very large negative dB values at high frequencies.
- Bottom subplot (time domain): original (black) overlaid with FIR (red) and IIR (blue). FIR output is visibly delayed but preserves waveform shape; IIR output shows reduced amplitude and some phase/shape differences (nonlinear phase and transient effects).

Key observations

- **FIR magnitude:**

- Linear-phase symmetry yields an approximately constant group delay $\approx N/2$ samples, visible as the time delay in the time-domain plot.
- Windowing sidelobes / spectral leakage produce periodic notches and ripples in the stopband.
- Passband is relatively flat and predictable.

- **IIR magnitude:**

- Provides steeper, monotonic attenuation for a given filter order, yielding stronger stop-band suppression with fewer coefficients.
- Lacks linear phase; phase varies with frequency, causing waveform distortion and altered transient behaviour.
- Extremely large negative dB values at high frequencies are visible (note: numerical flooring was likely applied to avoid $-\infty$ when plotting in dB).

- **Time domain:**

- FIR preserves waveform shape better, with the primary effect being a constant delay.
- IIR introduces amplitude/phase distortion and possible ringing around sharp events; overall output magnitude can be lower for this design.
- Both filters remove high-frequency content, but the FIR output aligns with the input except for delay, while the IIR output shows shape changes due to nonlinear phase.

Discussion

The results confirm the expectations: the windowed FIR (Hamming, order N) exhibits a smooth passband with low sidelobes and a transition width that narrows as N increases, while the Butterworth IIR (order M) attains a steeper roll-off at lower order; the FIR has essentially linear phase with constant group delay $\approx N/2$ preserving waveform shape, whereas the IIR shows nonlinear phase and may introduce transient ringing; numerically, magnitudes were floored before dB conversion to avoid $-\infty$ spikes and the IIR was obtained via the bilinear transform (prewarp when the cutoff is high relative to Nyquist); in practice choose FIR for phase fidelity and IIR for lower complexity/steeper attenuation, or compensate IIR phase / raise FIR order when required.

Conclusion

This experiment demonstrated the complementary strengths of windowed FIR and Butterworth IIR low-pass filters. The FIR (Hamming window) provides linear phase and predictable group delay at the expense of higher order to achieve a narrow transition band. The Butterworth IIR achieves sharper roll-off with lower order and computational cost but at the cost of nonlinear phase and potential transient distortion. Choice of filter should therefore be driven by application requirements: prefer FIR for phase-sensitive signals and IIR for resource-constrained designs needing rapid attenuation. Practical implementations must also account for numerical issues (dB flooring, coefficient quantization) and, for IIRs, pole stability.

References

- [1] A. V. Oppenheim and R. W. Schaffer, *Discrete-Time Signal Processing*, 2nd ed. Upper Saddle River, NJ: Prentice Hall, 1999.
- [2] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms and Applications*, 4th ed. Upper Saddle River, NJ: Prentice Hall, 2006.
- [3] R. G. Lyons, *Understanding Digital Signal Processing*, 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2011.